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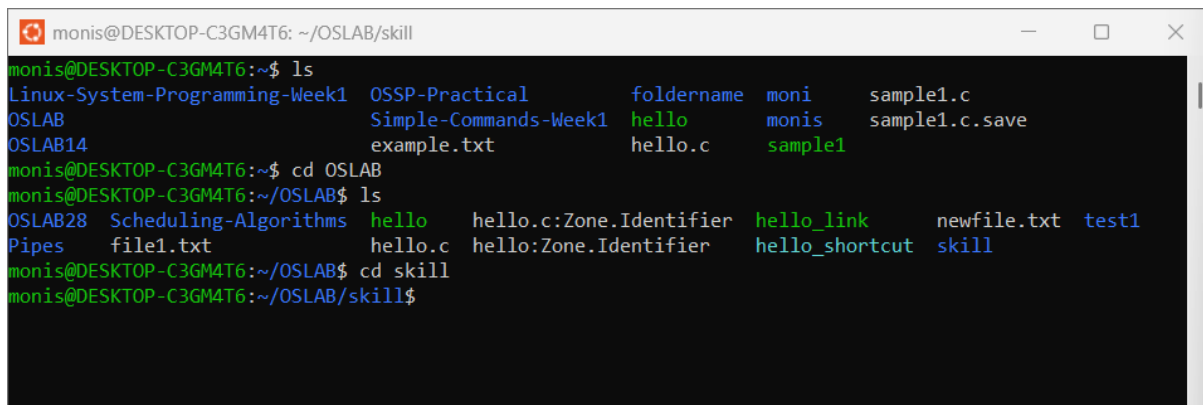
Section: 6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

7. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis   sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c hello:Zone.Identifier    hello_shortcut skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2 — Create the waitpid() Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 waitpid_monitor.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child process started\n");
        printf("Child PID: %d\n", getpid());

        sleep(2);

        printf("Child process completed\n");
        exit(0);
    }
    else
    {
        printf("Parent process started\n");
        printf("Parent PID: %d\n", getpid());

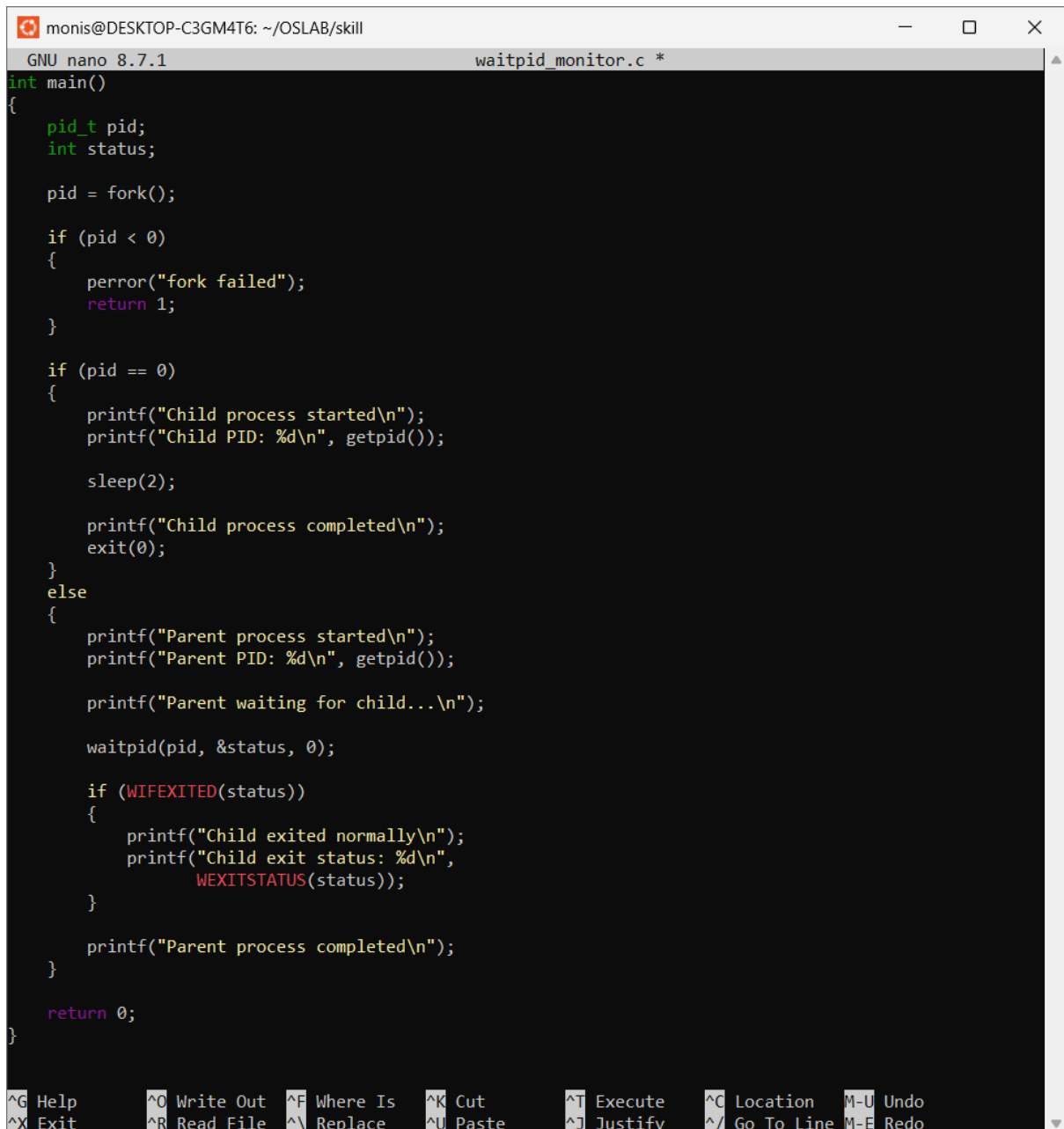
        printf("Parent waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
        {
            printf("Child exited normally\n");
            printf("Child exit status: %d\n",
                WEXITSTATUS(status));
        }

        printf("Parent process completed\n");
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^\\ Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 waitpid_monitor.c *
int main()
{
    pid_t pid;
    int status;

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child process started\n");
        printf("Child PID: %d\n", getpid());

        sleep(2);

        printf("Child process completed\n");
        exit(0);
    }
    else
    {
        printf("Parent process started\n");
        printf("Parent PID: %d\n", getpid());

        printf("Parent waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
        {
            printf("Child exited normally\n");
            printf("Child exit status: %d\n",
                WEXITSTATUS(status));
        }

        printf("Parent process completed\n");
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

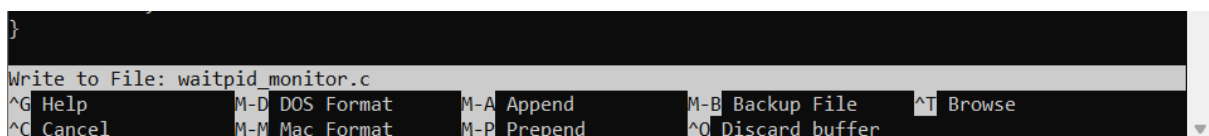
Save the file

Press:

Ctrl + O

Enter

Ctrl + X



```
}
Write to File: waitpid_monitor.c
^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^O Discard buffer
```

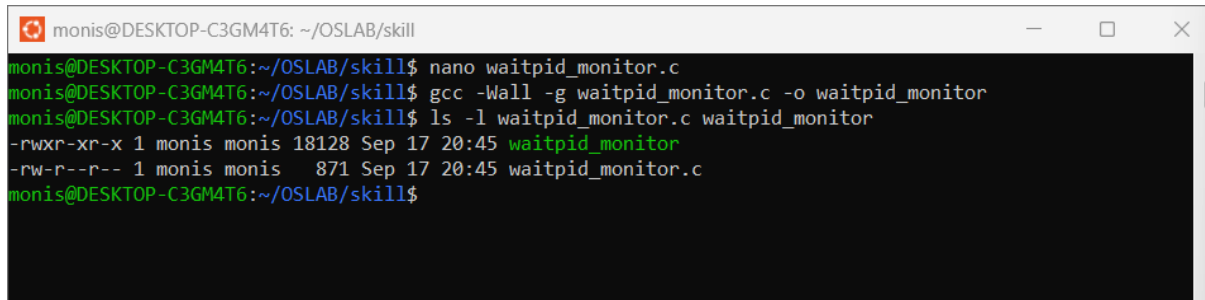
Step 3 — Compile the waitpid() Program

Run:

```
gcc -Wall -g waitpid_monitor.c -o waitpid_monitor
```

If there are no errors, check that the executable was created:

```
ls -l waitpid_monitor.c waitpid_monitor
```

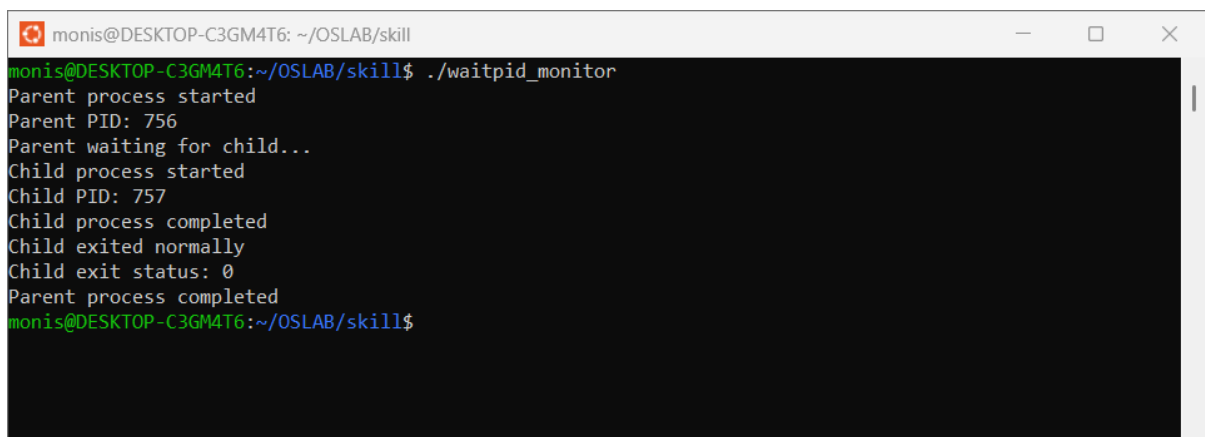
A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill' showing the compilation of waitpid_monitor.c. The user runs 'nano waitpid_monitor.c', then 'gcc -Wall -g waitpid_monitor.c -o waitpid_monitor', and finally 'ls -l waitpid_monitor.c waitpid_monitor'. The output shows the file permissions and details for both the source file and the executable.

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano waitpid_monitor.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g waitpid_monitor.c -o waitpid_monitor
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l waitpid_monitor.c waitpid_monitor
-rwxr-xr-x 1 monis monis 18128 Sep 17 20:45 waitpid_monitor
-rw-r--r-- 1 monis monis  871 Sep 17 20:45 waitpid_monitor.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 4 — Run the waitpid() Program

Now run:

```
./waitpid_monitor
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill' showing the execution of the waitpid_monitor program. The user runs './waitpid_monitor', and the output shows the parent process starting, waiting for a child, the child starting and completing, and the parent process completing.

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./waitpid_monitor
Parent process started
Parent PID: 756
Parent waiting for child...
Child process started
Child PID: 757
Child process completed
Child exited normally
Child exit status: 0
Parent process completed
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 5 — Retrieve the PATH Variable

Run:

```
echo $PATH
```

You should see something similar to:

```
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
Parent waiting for child...
Child process started
Child PID: 757
Child process completed
Child exited normally
Child exit status: 0
Parent process completed
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/ssl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.6.2.0_x64_0uekyb3d8bb
w:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/C
ommon Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows/system32/wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NV
IDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Win
dows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/Scripts:/mnt/c/Users/monis/AppData/Local
/Programs/Python/Python310:/mnt/c/Users/monis/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/monis/.local/bin:/mnt/c/Users/monis/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/monis/AppData/Local/Program
s/Microsoft VS Code/bin:/mnt/c/Users/monis/AppData/Roaming/npm:/mnt/c/Users/monis/AppData/Local/Python/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1.2/bin:/snap/bin
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Now run:
printenv PATH

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/ssl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.6.2.0_x64_0uekyb3d8bb
w:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/C
ommon Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows/system32/wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NV
IDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Win
dows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/Scripts:/mnt/c/Users/monis/AppData/Local
/Programs/Python/Python310:/mnt/c/Users/monis/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/monis/.local/bin:/mnt/c/Users/monis/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/monis/AppData/Local/Program
s/Microsoft VS Code/bin:/mnt/c/Users/monis/AppData/Roaming/npm:/mnt/c/Users/monis/AppData/Local/Python/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1.2/bin:/snap/bin
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ printenv PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/ssl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.6.2.0_x64_0uekyb3d8bb
w:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/C
ommon Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows/system32/wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NV
IDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Win
dows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/Scripts:/mnt/c/Users/monis/AppData/Local
/Programs/Python/Python310:/mnt/c/Users/monis/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/monis/.local/bin:/mnt/c/Users/monis/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/monis/AppData/Local/Program
s/Microsoft VS Code/bin:/mnt/c/Users/monis/AppData/Roaming/npm:/mnt/c/Users/monis/AppData/Local/Python/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1.2/bin:/snap/bin
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 6 — Create the PATH Resolution Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano path_resolution.c
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 path_resolution.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>

#define MAX_PATH 4096

int main()
{
    char *path;
    char *path_copy;
    char *directory;
    char full_path[MAX_PATH];
    struct stat file_info;

    path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH variable:\n%s\n", path);

    path_copy = strdup(path);

    if (path_copy == NULL)
    {
        perror("strdup");
        return 1;
    }

    printf("Search directories:\n");

    directory = strtok(path_copy, ":");

    while (directory != NULL)
    {
        printf("%s\n", directory);

        directory = strtok(NULL, ":");
    }

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 path_resolution.c *

    directory = strtok(NULL, ":");
}

free(path_copy);

printf("\nExecutable resolution:\n");

path_copy = strdup(path);

if (path_copy == NULL)
{
    perror("strdup");
    return 1;
}

directory = strtok(path_copy, ":");

while (directory != NULL)
{
    snprintf(full_path, sizeof(full_path),
             "%s/%s", directory, "ls");

    if (stat(full_path, &file_info) == 0)
    {
        if (access(full_path, X_OK) == 0)
        {
            printf("Found executable: %s\n", full_path);
            break;
        }
    }

    directory = strtok(NULL, ":");
}

free(path_copy);

if (directory == NULL)
{
    printf("Executable not found.\n");
}

return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo
```

Save the file

Press:

Ctrl + O

Enter

Ctrl + X

```
}
Write to File: path_resolution.c
^G Help      M-D DOS Format  M-A Append      M-B Backup File ^T Browse
^C Cancel     M-M Mac Format  M-P Prepend     ^O Discard buffer
```

Step 7 — Compile the PATH Resolution Program

Run:

```
gcc -Wall -g path_resolution.c -o path_resolution
```

If there are no errors, check that the source file and executable exist:

```
ls -l path_resolution.c path_resolution
```

[You should see both files listed.]

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano path_resolution.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g path_resolution.c -o path_resolution
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l path_resolution.c path_resolution
-rwxr-xr-x 1 monis monis 19192 Sep 17 20:55 path_resolution
-rw-r--r-- 1 monis monis 1479 Sep 17 20:55 path_resolution.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 8 — Run the PATH Resolution Program

Run:

```
./path_resolution
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./path_resolution
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/isl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.6.2.0_x64_8wekyb3d8b...
Search directories:
/usr/local/sbin
/usr/local/bin
/usr/sbin
/usr/bin
/sbin
/bin
/usr/games
/usr/local/games
/usr/lib/isl/lib
/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.6.2.0_x64_8wekyb3d8b...
/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.1-hotspot/bin
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Windows/system32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/OpenSSH/
/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common
/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NVOLISR
/mnt/c/Windows/System32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/OpenSSH/
/mnt/c/Program Files/dotnet/
/mnt/c/Program Files/Git/cmd
/mnt/c/Program Files/nodejs/
/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/Scripts/
/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/
/mnt/c/Users/monis/AppData/Local/Programs/Python/Launcher/
/mnt/c/Users/monis/AppData/Local/bin
/mnt/c/Users/monis/AppData/Local/Microsoft/WindowsApps
/mnt/c/Users/monis/AppData/Local/Programs/Microsoft VS Code/bin
/mnt/c/Users/monis/AppData/Roaming/npm
/mnt/c/Users/monis/AppData/Local/Python/bin
/mnt/c/Program Files/ JetBrains/PyCharm 2026.1.2/bin
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
/usr/local/sbin
/usr/local/bin
/usr/sbin
/usr/bin
/sbin
/bin
/usr/games
/usr/local/games
/usr/lib/ssl/lib
/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.6.2.0_x64_Bwekyb3d8bhue
/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.1.8-hotspot/bin
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Windows/system32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/PowerShell/
/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common
/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR
/mnt/c/Windows/System32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/PowerShell/
/mnt/c/Program Files/dotnet/
/mnt/c/Program Files/Git/cmd
/mnt/c/Program Files/nodejs/
/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/Scripts/
/mnt/c/Users/monis/AppData/Local/Programs/Python/Python310/
/mnt/c/Users/monis/AppData/Local/Programs/Python/Launcher/
/mnt/c/Users/monis/.local/bin
/mnt/c/Users/monis/AppData/Local/Microsoft/WindowsApps
/mnt/c/Users/monis/AppData/Local/Programs/Microsoft VS Code/bin
/mnt/c/Users/monis/AppData/Roaming/npm
/mnt/c/Users/monis/AppData/Local/Python/bin
/mnt/c/Program Files/JetBrains/PyCharm 2026.1.2/bin
/snap/bin

Executable resolution:
Found executable: /usr/bin/ls
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$
```

Step 9 — Test a Missing Command

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ nano path_missing.c
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 path_missing.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>

#define MAX_PATH 4096

int main()
{
    char *path;
    char *path_copy;
    char *directory;
    char full_path[MAX_PATH];
    struct stat file_info;
    int found = 0;

    path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    path_copy = strdup(path);

    if (path_copy == NULL)
    {
        perror("strdup");
        return 1;
    }

    directory = strtok(path_copy, ":");

    while (directory != NULL)
    {
        snprintf(full_path, sizeof(full_path),
                 "%s/%s", directory, "ossp_missing_command");

        if (stat(full_path, &file_info) == 0 &&
            access(full_path, X_OK) == 0)
        {
            printf("Found executable: %s\n", full_path);
            found = 1;
        }

        directory = strtok(NULL, ":");
    }

    return found ? 0 : 1;
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^\\ Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 path_missing.c *

path = getenv("PATH");

if (path == NULL)
{
    printf("PATH variable not found.\n");
    return 1;
}

path_copy = strdup(path);

if (path_copy == NULL)
{
    perror("strdup");
    return 1;
}

directory = strtok(path_copy, ":");

while (directory != NULL)
{
    snprintf(full_path, sizeof(full_path),
             "%s/%s", directory, "ossp_missing_command");

    if (stat(full_path, &file_info) == 0 &&
        access(full_path, X_OK) == 0)
    {
        printf("Found executable: %s\n", full_path);
        found = 1;
        break;
    }

    directory = strtok(NULL, ":");
}

free(path_copy);

if (!found)
{
    printf("Command not found in PATH.\n");
}

return 0;
}
```

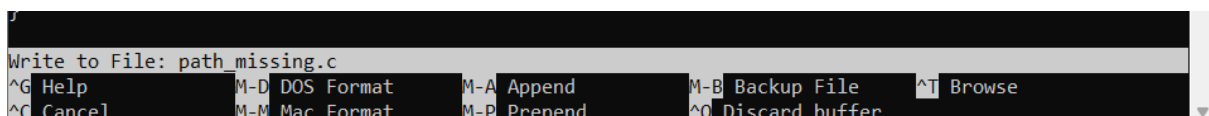
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

Save:

Ctrl + O

Enter

Ctrl + X



Write to File: path_missing.c

^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^O Discard buffer

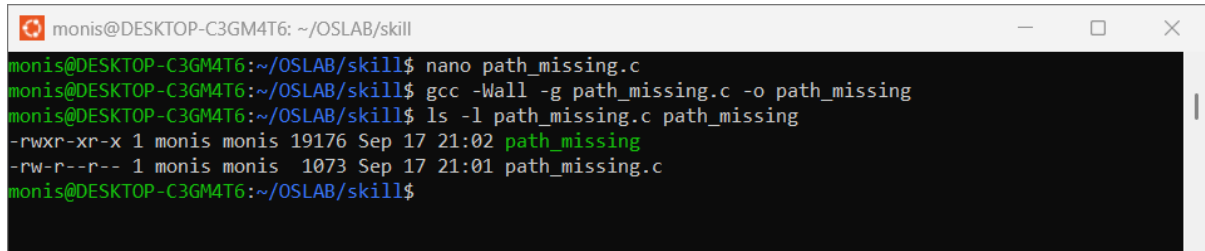
Step 10 — Compile the Missing-Command Test

Run:

```
gcc -Wall -g path_missing.c -o path_missing
```

If there are **no errors**, check that the files were created:

```
ls -l path_missing.c path_missing
```

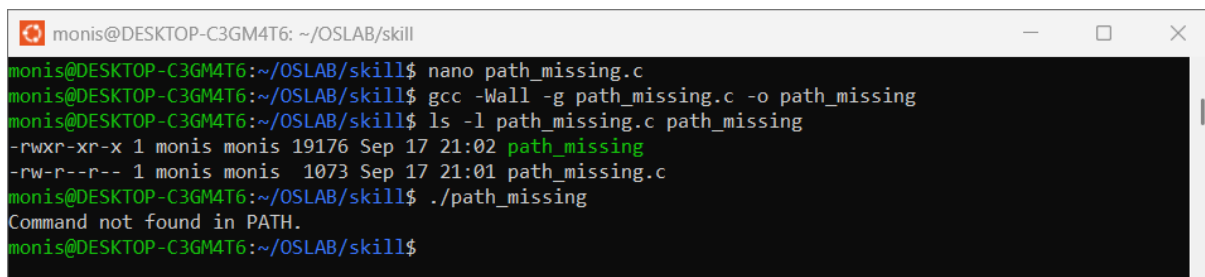


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano path_missing.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g path_missing.c -o path_missing
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l path_missing.c path_missing
-rwxr-xr-x 1 monis monis 19176 Sep 17 21:02 path_missing
-rw-r--r-- 1 monis monis 1073 Sep 17 21:01 path_missing.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 11 — Run the Missing-Command Test

Now run:

```
./path_missing
```

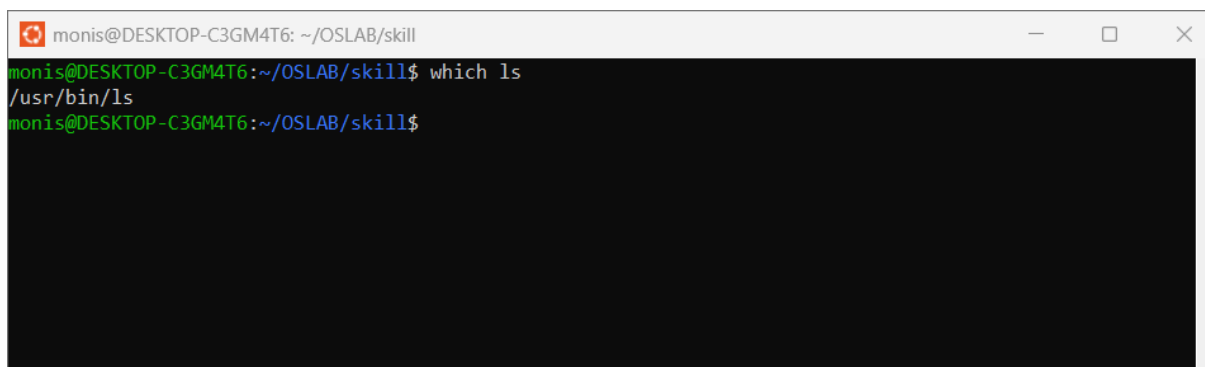


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano path_missing.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g path_missing.c -o path_missing
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l path_missing.c path_missing
-rwxr-xr-x 1 monis monis 19176 Sep 17 21:02 path_missing
-rw-r--r-- 1 monis monis 1073 Sep 17 21:01 path_missing.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./path_missing
Command not found in PATH.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 12 — Final PATH Resolution Test

Run:

which ls



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ which ls
/usr/bin/ls
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Then run:

```
ls -l $(which ls)
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ which ls
/usr/bin/ls
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l $(which ls)
lrwxrwxrwx 1 root root 29 Mar 30 22:20 /usr/bin/ls -> ../lib/cargo/bin/coreutils/ls
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```